SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	The primary objective of this project is to develop a **Machine Learning-based Credit Card Approval Prediction System** that automates the credit card approval process. The system analyzes applicant information and predicts whether a credit card application should be approved or rejected, enabling faster, more accurate, and consistent decision-making.
Scope	The project focuses on automating the credit card approval process by integrating machine learning with a user-friendly web application. It includes data preprocessing, model training, prediction generation, and a dashboard for users and administrators. The system aims to reduce manual effort, improve decision accuracy, and enhance the overall application review process.
Problem Statement	
Description	Banks receive a large number of credit card applications every day. Evaluating each application manually is time-consuming and prone to human errors. Factors such as income, employment status, credit history, and existing financial obligations must be analyzed before making an approval decision. Manual evaluation often results in inconsistent decisions and longer processing times.

Impact	Automating the approval process using machine learning will: • Reduce manual effort and processing time. • Improve the accuracy and consistency of approval decisions. • Minimize human errors. • Enhance customer satisfaction through faster responses. • Help financial institutions make better risk-based decisions.
Proposed Solution	
Approach	. The proposed solution uses a trained **Machine Learning classification model** to analyze applicant data and predict credit card approval eligibility. The model is integrated into a **Flask-based web application** that allows users to enter applicant information and receive instant prediction results.
Key Features	• Machine Learning-based credit card approval prediction. • Fast and accurate applicant eligibility assessment. • User-friendly web dashboard for data entry and predictions. • Data preprocessing and validation before prediction. • Real-time prediction results. • Model evaluation using Accuracy, Confusion Matrix, and Classification Report. • Responsive interface accessible from desktop and mobile devices.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing resources required to execute data preprocessing, model training, and prediction tasks efficiently.	Multi-core CPU (Intel Core i5/Ryzen 5 or higher).
Memory	Temporary memory required to run the Flask application, machine learning model, and data processing libraries simultaneously.	Minimum **8 GB RAM**

Storage	Storage space required for datasets, trained models, source code, reports, and project files.	Minimum **20 GB SSD** (or higher).
Software		
Frameworks	Software framework used to build the web application and integrate the machine learning model with the user interface	Flask.
Libraries	Libraries used for data preprocessing, model training, evaluation, visualization, and model serialization.	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib.
Development Environment	Tools used for writing, testing, debugging, and managing the application during development.	Visual Studio Code / PyCharm / Jupyter Notebook.
Data		
Data	Dataset characteristics required for model development and testing.	**CSV format**, approximately **690 applicant records.**